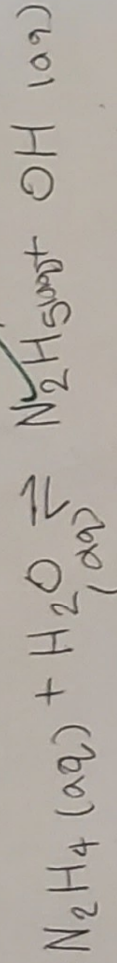


ANKING (10 MARKS)

If the K_b for hydrazine, $N_2H_4(aq)$, is 1.7×10^{-6} , calculate the pH of a 1.8 mol/L solution of hydrazine. (6 marks)



we can ignore

species are N_2H_4 and H_2O but $K_b \gg K_w$ so

contribution from water

	N_2H_4 1.8M	OH ⁻	$N_2H_5^+$
T		0	0
C	-x	+x	+x
E	1.8-x	x	x

$$[OH^-][N_2H_5^+] = \frac{(x)(x)}{(1.8-x)}$$

$$= \frac{(x)(x)}{(1.8-x)}$$

5% rule so (remove)

$$= x^2$$

$$x = 1.75 \times 10^{-3}$$

$$11.24$$

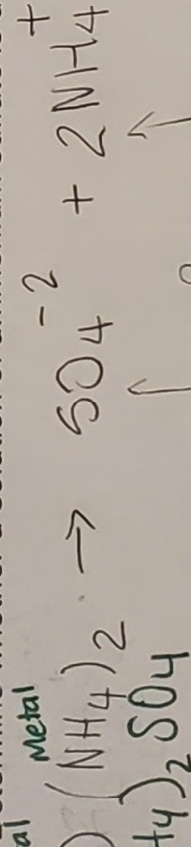
$$pH = 8.48$$

$$pOH = -\log [OH^-] = 5.514$$

calc error? 14 - 5.514 = 8.48

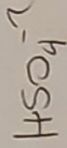
3.49, slightly basic.

Determine whether a solution of ammonium sulfate is acidic, basic, or neutral. Show all of your work! (4 marks)



conjugate of weak acid

conjugate of weak base



K_w

K_a for

$$HSO_4^- = 1 \times 10^{-2}$$

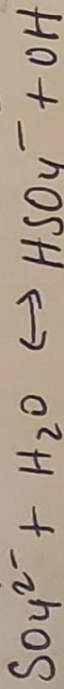
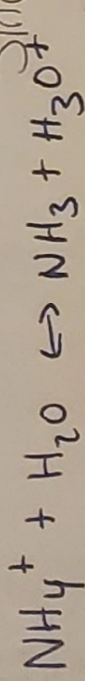
$$= \frac{1 \times 10^{-14}}{1 \times 10^{-2}}$$

$$= 1 \times 10^{-12}$$

show equations of above ions with water?

Since $K_a > K_b$ solution will be acidic.

Since



NH_4^+ is the conjugate of a weak base making a weak acid with

$$K_a = 5.8 \times 10^{-10}$$

SO_4^{2-} is the conjugate

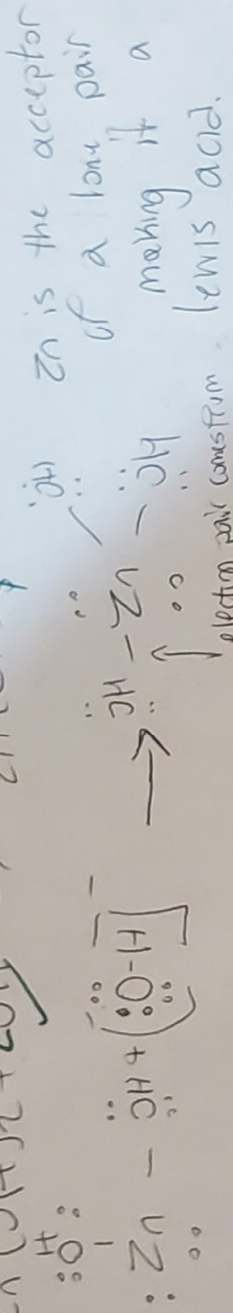
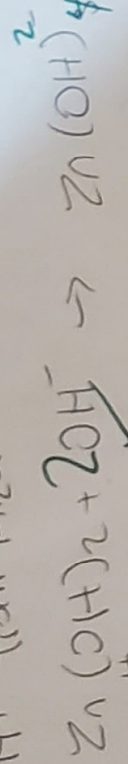
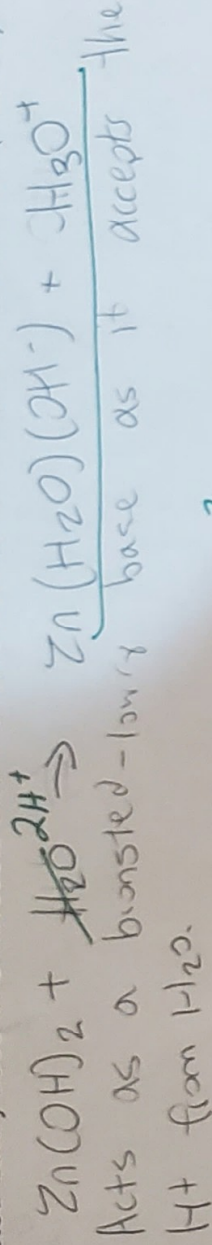
HSO_4^- (weak acid) making

a weak base. with $K_b =$

will be

COMMUNICATION (10 MARKS)

1. Zinc hydroxide is an amphoteric substance. Show, using equations for each, how $\text{Zn}(\text{OH})_2$ acts as a Bronsted-Lowry base when it reacts with H^+ and as a Lewis acid when it reacts with OH^- . (4 marks)



2. Place the species in each of the following groups in order of increasing acid strength. Give a reasoning for your answer. (4 marks)

a. H_2S , H_2O , H_2Se (bond energies: $\text{H}-\text{O}$, 467 kJ/mol; $\text{H}-\text{S}$, 363 kJ/mol; $\text{H}-\text{Se}$, 276 kJ/mol)

b. $\text{F}_3\text{CCO}_2\text{H}$, $\text{CH}_3\text{CO}_2\text{H}$, $\text{F}_2\text{CHCO}_2\text{H}$, $\text{FCH}_2\text{CO}_2\text{H}$

a) Strongest bond means wanting to keep H , thus less likely to donate

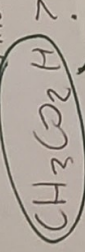
H_2Se Strongest

H_2S

H_2O Weakest

Using the difference between H and H_2O

b) And the more F 's are added the stronger the acid / look at



3. An antacid purchased at a local drug store has a pOH of 2.3. Calculate the pH , $[\text{H}^+]$, and $[\text{OH}^-]$ of this solution. Is the antacid acidic or basic? (2 marks)

$$\text{pOH} = 2.3$$

$$\text{pH} = 14 - 2.3 = 11.7$$

$$[\text{H}^+] = 10^{-\text{pH}} = 1.995 \times 10^{-12} \text{ M}$$

$$[\text{OH}^-] = 10^{-\text{pOH}} = 0.005 \text{ M}$$

The antacid is basic as the pH is high, meaning $[\text{H}^+]$ concentration is low, thus making it less acidic.